MINOR SPECIALIZATION RESEARCH PROJECT

Hybrid Market Predictor & Strategy Optimizer

LightGBM and NSGA-II for Indian stocks

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Project Purpose

- Learn patterns in historical stock data.
- Search for trading rules that balance return and risk.
- Test those rules on later, unseen data.

SYSTEM ARCHITECTURE & INPUTS

Data and Tools

- **NSE stocks:** Reliance, TCS, and Infosys.
- **Price and volume data:** Daily prices and trading volume via yfinance, saved as immutable snapshots.
- **Feature library:** 20 indicators describing market trend, momentum, and volatility.
- **Software stack:** Python with LightGBM, NSGA-II, and Streamlit for visual interaction.

EXECUTION PIPELINE

How It Works

1. Read historical daily market prices.
2. Build technical indicators describing trend and volatility.
3. LightGBM estimates the chance of a rise over the next 3 trading bars.
4. NSGA-II searches for entry, exit, and risk rules.
5. Backtest on later data and show results.

PREDICTIVE ML & EVOLUTIONARY SEARCH

The Two Algorithms

Tier 1: LightGBM

- Learns from past indicator examples.
- Produces a forward 3-day directional probability.

Directional Forecast: Estimates market probability rather than guaranteeing exact price levels.

Tier 2: NSGA-II

- Improves candidate trading rules over repeated rounds.
- Balances Sharpe ratio and maximum drawdown simultaneously.

Key Metrics:
- Sharpe = Return relative to volatility
- Drawdown = Largest peak-to-trough fall

6-YEAR ROLLING RETROSPECTIVE STUDY (2020–2025)

Research Results

Stock	Hybrid Return	Buy-and-Hold Return
Reliance (RELIANCE.NS)	-0.35%	128.74%
TCS (TCS.NS)	25.60%	65.13%
Infosys (INFY.NS)	9.10%	152.78%

Caption: 2020–2025 cumulative compounded return; hybrid strategy median across 3 random GA seeds.

- **Visible takeaway: Buy-and-hold earned more total return on all three stocks.**
- Defensive benefit: Hybrid maximum drawdown was lower in 17 of 18 annual stock comparisons.

Using the Dashboard

1. Select a stock or asset from the sidebar.
2. Set Candidate Population to 40, Improvement Rounds to 15, and Random Seed to 42.
3. Click the Run Experiment button.
4. Inspect performance KPIs, Pareto frontier strategies, and simulated trade logs.

Key Parameter Definitions:
Population = Candidate rules • Generations = Improvement rounds • Seed = Repeatable randomness

Note: Interactive dashboard executes a single chronological split for real-time exploration (separate from 6-year rolling research).

SUMMARY & FUTURE WORK

Conclusion and Next Steps

- Built an end-to-end prediction and strategy research tool.
- Observed risk-return trade-offs: lower drawdowns, but lower total return than buy-and-hold in this study.
- Next steps: evaluate broader stock universes and incorporate richer macroeconomic features.

Thank you for your time — Questions & Discussion